PixelRec: From Modality-Level Fusion to Signal-Patch-Level Fusion for Multimodal Sequential Recommendation

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Abstract

In Multimodal Sequential Recommendation (MMSR), many recent methods treat modality-level features as the basic fusion units. However, we argue that the fusion units are coarse-grained and we empirically find that their performance does not always achieve complementary gains from different modalities under masked-modality fusion analysis. Motivated by this, we revisit MMSR from a new perspective, shifting the basic fusion units from coarse-grained modality-level features to fine-grained signal-patch-level ones, and propose **PixelRec**. In PixelRec, textual content is first transformed into visualized text and then combined with the original image to form a unified visual input. This unified input that carries both visual and textual content is then encoded into signal patches by vision encoders. These signal patches are treated as the basic fusion units for the subsequent fusion. To adaptively fuse these signal patches into item representations, we further introduce a Vision-Recommendation Aggregator. Additionally, to address the high GPU cost associated with feature processing, we design a Re-Construction Compression module to reduce GPU consumption. Extensive experiments demonstrate that PixelRec outperforms state-of-the-art MMSR methods (e.g., PRISM, HM4SR, etc.), achieving improvements of 4.6%-15.9% in terms of recommendation accuracy.

CCS Concepts

• **Information systems → Recommender systems**.

Keywords

Recommender System, Vision Language Model, Multimodal Sequential Recommendation, Multimodal Fusion

1 Introduction

Sequential Recommendation (SR) aims to predict users' future interactions based on their historical interactions, and it is an important task in modern recommender systems [5, 14]. It has also become an important research direction in multimedia, where rich multimodal item content provides new opportunities for improving SR. To capture richer item representations, recent studies [2, 10] extend SR to Multimodal Sequential Recommendation (MMSR), which incorporates auxiliary modalities such as visual illustrations and textual descriptions. By leveraging signals from different modalities, MMSR methods have demonstrated superior capability in modeling user preferences and improving recommendation performance [8, 18, 34].

Existing MMSR methods have evolved along the direction of effectively integrating multimodal information. Notably, we find that many methods [18, 33, 34] follow a modality-level fusion paradigm. As illustrated in Figure 1(a), the fusion units in these methods are modality-level features: the input of each modality is first independently encoded by a text encoder (e.g., Large Language Models or CLIP [23]) or a vision encoder (ViTs [6] or CLIP), and the resulting modality-specific features are then used as the basic units for subsequent multimodal fusion. In other words, many current MMSR methods perform fusion between whole-modality representations.

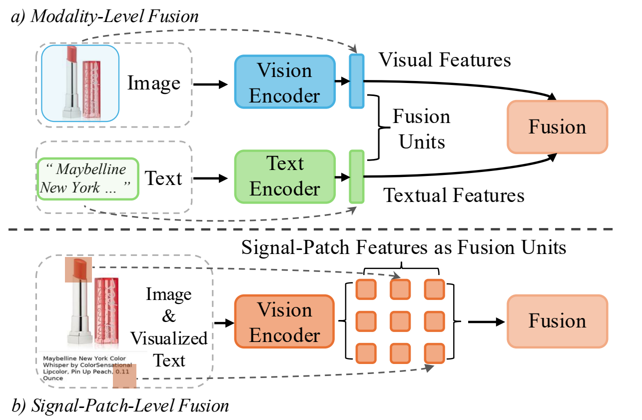

Figure 1: Comparison between modality-level fusion and signal-patch-level fusion. (a) Modality-level fusion treats the global representations of image and text as the basic fusion units. (b) Signal-patch-level fusion treats fine-grained signal patches extracted from a unified visual-textual input as the basic fusion units.

However, we find that such a design may suffer from a **modality compromise issue**: the final multimodal result may become a compromise between modalities, rather than a complementary gain from different modalities.

To investigate this phenomenon, we conduct a masked-modality fusion experiment on three datasets using three representative MMSR methods, where one modality is replaced with a blank input (e.g., empty image or text) while keeping the fusion process unchanged. Interestingly, the performance with full multimodal input is sometimes lower than that of the partially masked setting, where either the text modality is preserved and the image modality is masked, or the image modality is preserved and the text modality is masked (a detailed analysis is presented in Section 3.3). That is, the complete multimodal input does not always outperform the better masked setting under the same fusion framework. Based on recent findings in broader multimodal learning [9, 13], we consider that this limitation may stem from the current framework itself: the multimodal fusion units are coarse, making it difficult to stably transform heterogeneous signals into complementary gains.

In this work, we revisit MMSR from the perspective of fusion units and propose **PixelRec**. As Figure 1(b) shows, instead of using coarse-grained modality-level features as the basic fusion units, PixelRec performs multimodal fusion over fine-grained signal patches. Specifically, PixelRec no longer explicitly distinguishes whether a feature comes from the visual or textual modality. Instead, it treats each item's illustration and visualized text as a unified input and feeds them into the visual encoder of a Vision Language Model (VLM) to obtain signal-patch-level features. These signal-patch-level features then serve as the basic fusion units for subsequent fusion layers. To adaptively fuse the signal-patches, we introduce a Vision-Recommendation Aggregator (VRA), which leverages collaborative signals as the query to aggregate signal-patches into item representations. Furthermore, considering the high GPU cost of feature processing, we design a Re-Construction Compression (RCC) module to substantially reduce GPU memory consumption while preserving most of the recommendation performance.

Extensive experiments on multiple benchmarks demonstrate the effectiveness of PixelRec. Our method outperforms state-of-the-art MMSR approaches, achieving improvements of 4.6%-15.9%. These results validate the advantage of shifting the fusion units from a modality level to a signal-patch level for MMSR. Our contributions are as follows:

- We propose **PixelRec**, a new MMSR framework that revisits multimodal fusion from the perspective of fusion units, shifting the basic fusion units from coarse-grained modality-level features to fine-grained signal-patch-level ones. To the best of our knowledge, this unified visual-textual fusion method remains underexplored in MMSR.
- We design a Vision-Recommendation Aggregator (VRA) to adaptively aggregate recommendation-relevant signal patches, and introduce a Re-Construction Compression (RCC) module to reduce GPU memory consumption.
- Extensive experiments on datasets show that PixelRec outperforms SOTA MMSR baselines, validating the effectiveness of signal-patch-level fusion for MMSR.

2 Related Work

Multimodal Sequential Recommendation. MMSR extends conventional SR by incorporating auxiliary item content such as images and text to enrich item semantics. Existing MMSR methods typically follow a standard pipeline in which image and text are first encoded by modality-specific encoders and then fused for downstream sequence modeling [11, 27, 33]. Recent studies mainly improve this paradigm through adaptive fusion, dynamic fusion, and mixture-of-experts style interaction over multimodal features [18, 34]. Different from these methods, our work does not further refine modality-level fusion. Instead, we revisit MMSR from a finer-grained perspective and shift the modeling focus from whole-modality representations to signal-patch-level representations.

Vision Language Model. Vision-Language Models (VLMs) learn multimodal representations by aligning visual content with language supervision at scale [17, 28, 32]. Compared with conventional visual encoders such as ViTs [6], VLMs are better at capturing cross-modal correspondence and understanding visually embedded text in a unified representation space [4, 17]. Many VLMs adopt ViT-style visual encoders, whose patch-based architecture preserves spatial correspondence between output tokens and input regions, making them suitable for encoding unified visual-textual inputs.

Unified Visual-Textual Representation. Recent studies in OCR-free document understanding and visually situated language understanding have shown that text can be modeled as part of the visual input, rather than processed through an independent language branch [3, 15, 31]. Representative works such as Donut, Pix2Struct, and UReader demonstrate that visual content and embedded text can be jointly encoded in a unified visual-textual representation space. Different from these works, our goal is not OCR-free reading or document understanding, but recommendation-oriented multimodal modeling. To the best of our knowledge, this unified visual-textual paradigm has not been explored in MMSR. Our work introduces it into SR by transforming textual content into image-form input and jointly encoding it with the item image, thereby enabling signal-patch-level multimodal modeling.

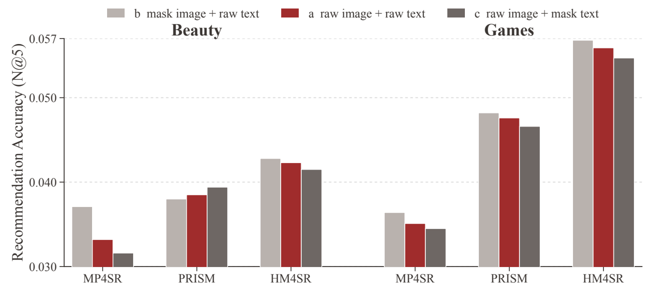

Figure 2: Masked-modality experiment results of baseline methods under different image-text settings (raw vs. masked).

3 Preliminary

3.1 Modality-Level Fusion in MMSR

Let $\mathcal{S}_u = [i_1, i_2, \ldots, i_T]$ denote the historical interaction sequence of user u, where the goal of sequential recommendation is to predict the next item that the user is likely to interact with. In MMSR, each item i is associated with multimodal content, including an image x_i^{img} and a textual title x_i^{txt}. Many MMSR methods organize item modeling around four components, e.g., a visual encoder, a text encoder, a multimodal fusion module, and a recommendation network. The mainstream pipeline is often to first encode each modality separately and then fuse them for downstream sequential modeling. Formally, the basic fusion units, e_i^{img} and e_i^{txt}, can be summarized as follows:

$$\mathbf{e}_i^{img} = f_{img}(x_i^{img}), \quad \mathbf{e}_i^{txt} = f_{txt}(x_i^{txt}) \tag{1}$$

where $f_{img}(\cdot)$ and $f_{txt}(\cdot)$ denote the modality-specific encoders. This paradigm is widely adopted in MMSR because it offers a simple and modular way to exploit visual and textual signals.

3.2 Masked-Modality Fusion Analysis

Previous studies [18, 34] typically remove multimodal inputs and corresponding fusion structures to simplify the model into a single-modal one for verifying the importance of multimodal content. Although such practices confirm the necessity of multimodal information, the actual role of multimodal fusion mechanisms remains under-explored.To better characterize fusion behaviors in MMSR, we adopt masked-modality fusion analysis [16, 20]. Different from these prior ablation methods, this setting preserves the original multimodal architecture and fusion pathway, and masks one modality throughout both training and evaluation. Thus, it probes modality interaction under the original multimodal fusion mechanism rather than the standalone ability of a single-modality variant.

Specifically, we consider three input variants in Figure 2: (a) *full multimodal* (*raw image + raw text*); (b) *text preserved + image masked* (*mask image + raw text*), where the image is replaced by an all-white blank image; and (c) *image preserved + text masked* (*raw image + mask text*), where the text is replaced by a blank string consisting only of spaces (' '). For all variants, we report recommendation accuracy in terms of *N@5*.

Ideally, the full multimodal variant should outperform the better one of the two masked variants, since the masked modality is only a format-preserving but uninformative signal. However, Figure 2 shows that this is not observed in many MMSR models with modality-level fusion units (e.g., MP4SR, PRISM, and HM4SR). In some cases, replacing one modality with such an information-free input even improves performance. This suggests that existing modality-level fusion mechanisms may suffer from a *compromise* effect, where multimodal interaction does not produce complementary gains and may even dilute useful signals.

3.3 Rethinking the Fusion Units in MMSR

The above phenomenon can be understood from two perspectives. First, separate encoding naturally produces modality representations with unequal predictive strengths before fusion. Since image and text are encoded by different encoders and summarized into independent modality-level embeddings, their downstream utility is not guaranteed to be balanced. Prior multimodal studies [22, 29] have shown that predictive dominance across modalities is common, and the dominant modality can suppress the optimization and effective utilization of other modalities during joint learning. Our masked-modality fusion analysis further confirms that such imbalance is also present in MMSR.

Second, many conventional multimodal systems rely on late fusion of final modality representations, such that cross-modal interaction is mainly performed at a coarse modality level [21]. Related multimodal studies have shown that all-at-once or coarse-grained fusion can overlook fine-grained local correlations [9, 13], while effective cross-modal matching often depends on fine-grained alignment between local visual regions and textual units [35]. Moreover, joint multimodal training may induce modality competition, causing stronger modalities to dominate and weaker ones to be under-utilized [7, 12].

Therefore, the core limitation of many MMSR methods may not be the lack of a better modality fusion method. Instead, we argue that the limitation might be that the fusion unit itself is too coarse.

4 Method

4.1 Framework Overview

Motivated by the findings and analysis in Section 3.3, we revisit the fusion units of MMSR from a coarse modality-level to a more fine-grained level, which leads to our proposed PixelRec. As illustrated in Figure 3, PixelRec models multimodal items through four stages: unified visual-textual rendering, VLM-based token extraction, RCC-based feature compression, and VRA-based aggregation. Instead of encoding these modalities separately and fusing them at the modality level, PixelRec first transforms the textual title into image-form input and combines it with the original item image to construct a unified visual-textual representation. Formally, the unified visual input is defined as

$$\mathbf{v}_i = \mathcal{R}(x_i^{img}, x_i^{txt}), \tag{2}$$

where $\mathcal{R}(\cdot)$ denotes the visual-textual rendering operation.

The rendered input is then fed into the visual encoder of a VLM, which produces a set of spatial token representations:

$$\mathbf{Z}_i = \mathrm{VLM}(\mathbf{v}_i) = [\mathbf{z}_{i,1}, \dots, \mathbf{z}_{i,N}], \qquad \mathbf{Z}_i \in \mathbb{R}^{N\times d_v}, \tag{3}$$

where N is the number of native VLM tokens and d_v is the token feature dimension.

Since the native VLM output contains a relatively large number of tokens, we further downsample the token sequence by local average pooling to obtain a compact set of signal patches:

$$\mathbf{P}_i = \mathrm{Pool}(\mathbf{Z}_i) = [\mathbf{p}_{i,1}, \dots, \mathbf{p}_{i,M}], \qquad \mathbf{P}_i \in \mathbb{R}^{M\times d_v}, \tag{4}$$

where $\mathrm{Pool}(\cdot)$ denotes local average pooling and M is the number of retained signal patches. These signal patches serve as the fine-grained multimodal units of PixelRec.

Although downsampling reduces the number of signals, each signal patch still remains high-dimensional. Therefore, PixelRec further applies Re-Construction Compression (RCC) to compress patch-signal features. After RCC, each item is represented by a compact set of compressed signal patches, which serves as its raw multimodal embedding.

During downstream SR tasks, Vision Recommendation Aggregation (VRA) is applied to adaptively fuse different patches in the multimodal embedding into the final item representation $\mathbf{e}_i$ with collaborative signals (item ID embeddings). The resulting item representation is then used for sequence modeling.

In our implementation, the VLM is kept frozen and used only as a feature extractor. The extracted signal patches are compressed by RCC and cached offline, and the cached compressed features are used as inputs for downstream recommendation training. Furthermore, feature extraction in PixelRec is performed only through the visual encoder of the VLM. The whole process is OCR-free: the textual title is not converted into text tokens again, but is directly encoded as visual signals together with the item image. Although prompt-conditioned feature extraction is also feasible, we use prompt-free visual feature extraction in the final setting.

4.2 Re-Construction Compression

The top-right part of Figure 3 corresponds to the RCC module. After VLM extraction, patch-signal features are item-level representations and are independent of user sequences. In MMSR, such item features are commonly cached offline for downstream recommendation training. However, if the native VLM features are directly cached and fed into the recommendation model, their high feature dimensionality still introduces high GPU memory usages. To address this issue, we propose Re-Construction Compression (RCC) to compress sequence-independent patch-signal features along the feature dimension while preserving their semantic information.

For a patch-signal representation $\mathbf{p}_{i,m} \in \mathbb{R}^{d_v}$, RCC performs linear compression and reconstruction as

$$\tilde{\mathbf{p}}_{i,m} = W_{enc}\mathbf{p}_{i,m}, \qquad \hat{\mathbf{p}}_{i,m} = W_{dec}\tilde{\mathbf{p}}_{i,m}, \tag{5}$$

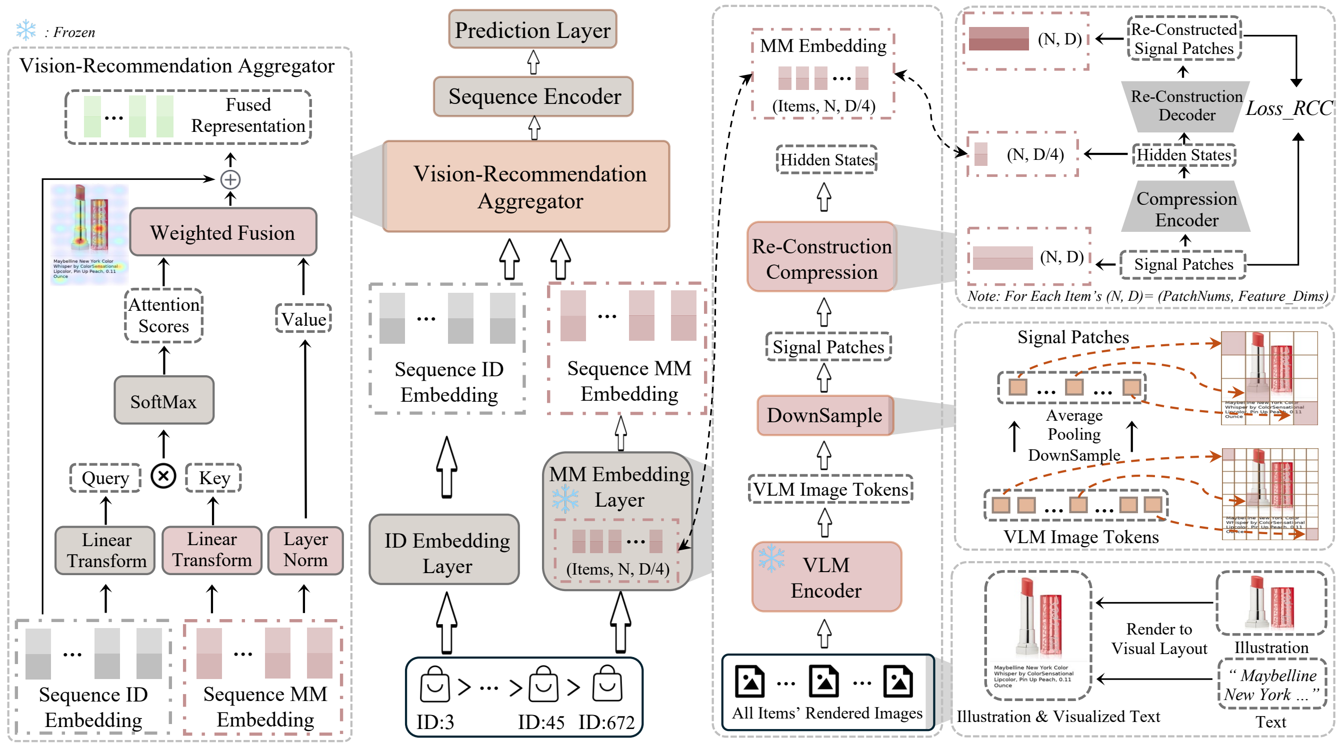

Figure 3: The workflow of the proposed PixelRec.

where $W_{enc} \in \mathbb{R}^{d_c \times d_v}$ and $W_{dec} \in \mathbb{R}^{d_v \times d_c}$ denote the compression and reconstruction matrices, respectively, and $d_c < d_v$ is the bottleneck dimension. After RCC, the compressed patch-signal representation of item i is written as

$$\tilde{\mathbf{P}}_i = [\tilde{\mathbf{p}}_{i,1}, \dots, \tilde{\mathbf{p}}_{i,M}] \in \mathbb{R}^{M \times d_c}. \tag{6}$$

To preserve the semantics of compressed patch-signal features, RCC is trained with a reconstruction objective consisting of a Smooth L1 loss and a cosine consistency term:

$$\mathcal{L}_{rcc} = \frac{1}{|\mathcal{B}|} \sum_{\mathbf{p} \in \mathcal{B}} \left[\mathrm{SmoothL1}(\hat{\mathbf{p}}, \mathbf{p}) + (1 - \cos(\hat{\mathbf{p}}, \mathbf{p})) \right], \tag{7}$$

where $\mathcal{B}$ denotes a batch of sampled signal patches.

In practice, RCC is trained offline on downsampled patch-signal representations, and the compressed patch cache is then used in downstream recommendation training. This design reduces the memory footprint of cached item-level features while preserving the recommendation utility of patch-level representations.

4.3 Vision-Recommendation Aggregator

The left part of Figure 3 corresponds to the VRA module. VRA operates on RCC-compressed signal patches and performs recommendation-oriented patch aggregation. Compared with modality-level fusion, such patch-level modeling preserves finer local information from the unified visual-textual input. However, not all signal patches contribute equally to recommendation. To identify the signals that are useful for recommendation, we design the Vision-Recommendation Aggregator, which adaptively aggregates item-specific informative patches into the final item representation.

Specifically, VRA maintains a set of learnable query vectors to attend over compressed signal patches. In the transductive setting, these queries are further conditioned on the item ID embedding, so that patch selection can be guided by collaborative signals. Formally, the query matrix is defined as

$$\mathbf{Q}_i = \mathbf{Q}_0 + \mathbf{1}_K (W_c \mathbf{e}_i^{id})^\top, \tag{8}$$

where $\mathbf{Q}_0 \in \mathbb{R}^{K \times d_a}$ is the matrix of learnable base queries, $\mathbf{e}_i^{id}$ is the item ID embedding, W_c projects the ID embedding into the query space, and $\mathbf{1}_K \in \mathbb{R}^K$ is an all-one vector used for broadcasting.

We then compute the projected keys and values as

$$\mathbf{K}_i = \tilde{\mathbf{P}}_i W_k, \qquad \mathbf{V}_i = \mathrm{LN}(\tilde{\mathbf{P}}_i), \tag{9}$$

where $W_k \in \mathbb{R}^{d_c \times d_a}$ projects compressed patch-signal features into the attention space and $\mathrm{LN}(\cdot)$ denotes layer normalization. The attention weights are computed by

$$\mathbf{A}_i = \mathrm{Softmax}\left(\frac{\mathbf{Q}_i \mathbf{K}_i^\top}{\sqrt{d_a}} \right), \tag{10}$$

and the query-specific patch responses are aggregated as

$$\mathbf{H}_i = \mathbf{A}_i \mathbf{V}_i. \tag{11}$$

Each query is expected to capture one type of recommendation-relevant local pattern. To further combine these multi-query responses, we compute a query-level importance score and perform weighted fusion:

$$\alpha_{i,k} = \frac{\exp(\mathbf{w}_s^\top \mathbf{H}_{i,k})}{\sum_{k'=1}^{K} \exp(\mathbf{w}_s^\top \mathbf{H}_{i,k'})}, \qquad \mathbf{h}_i = \sum_{k=1}^{K} \alpha_{i,k} \mathbf{H}_{i,k}, \tag{12}$$

where $\mathbf{w}_s \in \mathbb{R}^{d_c}$ is the query scoring vector, $\alpha_{i,k}$ denotes the importance weight of the k-th query response, and $\mathbf{H}_{i,k}$ is the k-th row of $\mathbf{H}_i$. Finally, the aggregated feature is projected into the recommendation space to form the item representation:

$$\mathbf{e}_i = \mathrm{Proj}(\mathbf{h}_i), \tag{13}$$

where $\mathrm{Proj}(\cdot)$ denotes the output projection module.

In this way, VRA shifts multimodal item modeling from modality-level weighting to patch-level weighting. Instead of deciding which modality is more important as a whole, PixelRec directly identifies which signal patches are more informative for recommendation.

5 Experiment

In this section, we investigate the following questions: **RQ1:** How does PixelRec perform compared with SOTA methods? **RQ2:** Can PixelRec alleviate the compromised performance issue observed in masked-modality experiments? **RQ3:** How does each component contribute to PixelRec's performance? **RQ4:** What does the fine-grained signal-patch-level modeling learned by PixelRec look like in visualization? **RQ5:** How do the different sampling strategies, RCC dims and prompt settings affect PixelRec's performance?

5.1 Experimental Setup

Datasets: Following prior MMSR studies, we conduct experiments on three widely used datasets from Amazon, e.g., *Beauty* [24, 25], *Games* [34] (short for *Video Games*), and *Toys* [34] (short for *Toys and Games*). Statistics about datasets are presented in Table 1. Following prior MMSR studies, we adopt the 5-core version of each dataset, where each user and each item has at least five interactions.
Evaluation: To ensure fair evaluation, we follow the standard leave-one-out protocol widely adopted in SR. For each user sequence, the last interacted item is used for testing, the second-to-last item is used for validation, and all preceding items are used for training. We report two widely used ranking metrics, HitRate@K (H@K) and NDCG@K (N@K), with $K \in \{5, 10, 20\}$ HitRate@K measures whether the ground-truth item appears in the top-K recommendation list, while NDCG@K further considers the ranking position of the target item. Higher values of both metrics indicate better recommendation performance.
Baselines: We compare PixelRec with two categories of baselines. **1) state-of-the-art traditional SR methods**: SASRec [14] from ICDM'18, BERT4Rec [26] from CIKM'19, BSARec [25] from AAAI'24, and TV-Rec [24] from NeurIPS'25; and **2) state-of-the-art MMSR methods**: UniSRec [10] from KDD'22, MP4SR [33] from RecSys'24, HM4SR [34] from WWW'25, IISAN-Versa [8] from TKDE'25 and PRISM [18], to appear in WWW'26. For PRISM, the official implementation releases the PRISM+STOSA and PRISM+SASRec variants. Following its released setting, we report the stronger PRISM+STOSA variant in our comparisons.
Implementation: All models are implemented in PyTorch. For a fair comparison, each baseline is configured with the best hyper-parameters reported in its original paper or further tuned by grid search. Following prior studies, we set the training batch size as 256 and the maximum length of each sequence is limited to 50. For PixelRec, we adopt Qwen3-VLM-8B [1] as our VLM encoder. For hyper-parameters, the downsampling patch nums are selected from {32 (8 ×4), 16 (4 × 4), 8 (4 × 2)}; the RCC output token dims d_c are selected from {4096, 1024, 256, 64}. Other details about the hyper-parmeter settings and rendering settings are presented in Appendix. All experiments are conducted on NVIDIA L20 GPUs. Codes and training logs are included in the supplementary materials.

Table 1: Dataset statistics.

Statistics	Beauty	Games	Toys
#Users	22,363	24,303	19,412
#Items	12,101	10,672	11,924
#Interactions	198,502	207,477	167,597
#Sparsity	99.93%	99.91%	99.93%
#Average Sequence Length	8.9	9.5	8.6

5.2 Overall Performance (RQ1)

Based on the results in Table 2, we have the following observations: (1) compared with the second-best results, the gains of PixelRec on H@5 and N@5 are generally more pronounced than those on H@20 and N@20. This indicates that PixelRec not only improves overall recall, but more importantly ranks the correct items at higher positions in the recommendation list. (2) SOTA non-multimodal SR methods remain highly competitive. Although the strongest multimodal baselines perform well on Beauty and Toys, SOTA non-multimodal models still achieve strong results on Games. This suggests that simply incorporating multimodal information does not necessarily guarantee better recommendation performance. The key lies in whether multimodal signals can be effectively integrated with sequential preference modeling.

The results of MMSR baselines reported in Table 2 follow the feature extractors used in the original papers. To further eliminate the influence of feature extractors, we use the same Qwen3-VLM-8B employed in PixelRec to separately encode the textual and visual contents. Considering the dimensional mismatch between the original networks and VLM features, we adopt RCC for dimension reduction to adapt to their original settings. Based on results in Table 3, we have the following observations: (1) PixelRec still outperforms baselines equipped with the same VLM feature extractor, indicating that its improvement cannot be attributed solely to a stronger VLM. (2) after replacing the original feature extractors with a VLM, the baseline performance may either improve or decline, showing that a stronger pretrained multimodal encoder alone is not sufficient to guarantee better recommendation performance.

5.3 Masked-Modality Fusion Analysis (RQ2)

In this section, we verify the ability of PixelRec to alleviate the modality compromise issue described in Section 3.3. This issue describes that many recent MMSR methods may suffer from a *modality compromise* effect under fixed multimodal fusion architectures, where multimodal fusion does not consistently translate heterogeneous inputs into complementary gains. To ensure the

Table 2: Main recommendation results on three datasets. Best results are shown in bold and second-best results are underlined. All improvements are statistically significant, as determined by a paired t-test with $p < 0.05$.

Method	Beauty						Games						Toys					
	H@5	H@10	H@20	N@5	N@10	N@20	H@5	H@10	H@20	N@5	N@10	N@20	H@5	H@10	H@20	N@5	N@10	N@20
Non-MM																		
SASRec [ICDM'18]	0.0382	0.0594	0.0877	0.0245	0.0313	0.0384	0.0576	0.0937	0.1445	0.0358	0.0474	0.0602	0.0454	0.0681	0.0944	0.0314	0.0387	0.0454
BERT4Rec [CIKM'19]	0.0463	0.0707	0.1047	0.0308	0.0387	0.0472	0.0776	0.1222	0.1857	0.0510	0.0653	0.0813	0.0468	0.0708	0.1030	0.0321	0.0398	0.0479
BSARec [AAAI'24]	0.0695	0.0992	0.1319	0.0488	0.0587	0.0677	0.0938	0.1420	0.2064	0.0617	0.0782	0.0929	0.0808	0.1065	0.1401	0.0571	0.0670	0.0755
TVRec [NeurIPS'25]	0.0724	0.0998	0.1373	0.0511	0.0599	0.0694	0.0943	0.1433	0.2090	0.0628	0.0785	0.0950	0.0807	0.1072	0.1438	0.0575	0.0661	0.0753
MM Models																		
UniSRec [KDD'22]	0.0610	0.0956	0.1415	0.0380	0.0491	0.0606	0.0723	0.1215	0.1894	0.0465	0.0624	0.0795	0.0681	0.1034	0.1440	0.0399	0.0513	0.0616
MP4SR [RecSys'24]	0.0601	0.0942	0.1401	0.0332	0.0442	0.0557	0.0638	0.1088	0.1703	0.0351	0.0495	0.0650	0.0770	0.1191	0.1652	0.0403	0.0539	0.0655
HM4SR [WWW'25]	0.0583	0.0804	0.1091	0.0423	0.0496	0.0564	0.0807	0.1190	0.1764	0.0559	0.0692	0.0811	0.0667	0.0889	0.1160	0.0501	0.0570	0.0634
IISAN-Versa [TKDE'25]	0.0733	0.1059	0.1475	0.0511	0.0616	0.0721	0.0509	0.0764	0.1090	0.0431	0.0593	0.0513	0.0814	0.1185	0.1601	0.0552	0.0672	0.0777
PRISM-STOSA [WWW'26]	0.0561	0.0765	0.1087	0.0385	0.0483	0.0552	0.0713	0.1075	0.1579	0.0476	0.0593	0.0719	0.0623	0.0823	0.1084	0.0465	0.0530	0.0595
PixelRec	**0.0850**	**0.1194**	**0.1647**	**0.0593**	**0.0703**	**0.0817**	**0.0994**	**0.1499**	**0.2210**	**0.0664**	**0.0825**	**0.1004**	**0.0909**	**0.1274**	**0.1736**	**0.0628**	**0.0746**	**0.0863**
Improvement	15.96%	12.75%	11.66%	16.05%	14.12%	13.31%	5.41%	4.61%	5.74%	5.73%	5.10%	5.68%	11.67%	6.97%	5.08%	9.22%	11.01%	11.07%

Table 3: N@5 results of VLM-enhanced multimodal baselines on three datasets. Δ denotes the relative change over the original settings in Table 2; $\uparrow$ indicates improvement and $\downarrow$ indicates degradation.

Method	Beauty	Δ	Games	Δ	Toys	Δ
UniSRec[w/ VLM]	0.0395	$\uparrow$	0.0412	$\downarrow$	0.0401	$\uparrow$
MP4SR[w/ VLM]	0.0336	$\uparrow$	0.0366	$\uparrow$	0.0349	$\downarrow$
HM4SR[w/ VLM]	0.0403	$\downarrow$	0.0523	$\downarrow$	0.0469	$\downarrow$
IISAN-Versa[w/ VLM]	0.0436	$\downarrow$	0.0304	$\downarrow$	0.0485	$\downarrow$
PRISM-STOSA[w/ VLM]	0.0397	$\uparrow$	0.0458	$\downarrow$	0.0473	$\uparrow$

consistency and comparability of experimental results, we adopt the same experimental method as that used in Section 3.3.

As presented in Figure 4, we observe that PixelRec achieves complementary gains under full multimodal input across all evaluated datasets. This indicates that the signal-patch-level modeling of PixelRec can indeed alleviate the modality compromise phenomenon.

5.4 Ablation Study (RQ3)

To better understand the contribution of each design choice in PixelRec, we conduct an ablation study from the perspective of multimodal integration strategy. As shown in Table 4, *Input* denotes the modality configuration, where *T*, *V*, and *T* / *V* represent textual input, visual input, and their joint input, respectively. *Unit* indicates the integration unit, i.e., *Modality* or *Patch*. *Fusion* denotes

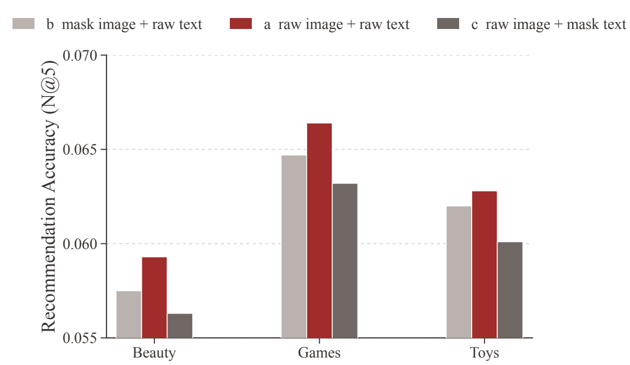

Figure 4: Masked-modality fusion experiment results of PixelRec on Beauty, Games and Toys datasets.

the aggregation strategy, including *Gated* and *VRA*. *ID* indicates whether ID-guided query attention is used in VRA. The metric Δ reports the relative improvement over the *Backbone* setting. Based on Table 4, we have the following observations: **(1) patch-level modeling is more effective than modality-level modeling.** Replacing modality-level units with patch-level units consistently improves the results under the same multimodal input and gated fusion, showing that finer-grained signal patches are better fusion units than whole-modality features. **(2) VRA further improves patch-level modeling.** Compared with gated fusion, VRA achieves

Table 4: Ablation study of multimodal integration strategies. Metrics: N@5 and Δ (% improvement over Backbone).

Input	Unit	Fusion	ID	Beauty		Games		Toys	
				N@5	Δ	N@5	Δ	N@5	Δ
–	–	–	–	0.0467	–	0.0613	–	0.0568	–
T	–	–	–	0.0528	+13.1%	0.0659	+7.5%	0.0604	+6.3%
V	–	–	–	0.0530	+13.5%	0.0648	+5.7%	0.0590	+3.9%
T / V	Modality	Gated	No	0.0537	+15.0%	0.0652	+6.4%	0.0607	+6.9%
T / V	Patch	Gated	No	0.0561	+20.1%	0.0648	+5.7%	0.0612	+7.7%
T / V	Patch	VRA	No	0.0575	+23.1%	0.0654	+6.7%	0.0619	+9.0%
T / V	Patch	VRA	Yes	**0.0593**	**+27.0%**	**0.0664**	**+8.3%**	**0.0628**	**+10.6%**

Table 5: Comparison between separate and unified visual-textual encoding. Results are reported in terms of N@5.

Input	Beauty	Games	Toys
Separate T/V	0.0571	0.0652	0.0614
Unified T/V	**0.0593**	**0.0664**	**0.0628**

consistently better results under the same patch-level setting, indicating the importance of recommendation-oriented patch aggregation. **(3) ID guidance brings additional gains on top of VRA.** Enabling ID-guided query attention consistently improves the results, suggesting that item-specific guidance helps the model focus on more relevant signal patches for different items. Overall, the results verify that the effectiveness of PixelRec comes from a progressive design: patch-level fusion provides the main gain, while VRA and ID guidance further improve recommendation-oriented multimodal modeling.

Moreover, to verify the necessity of unified visual-textual encoding, we conduct an additional experiment without visual-textual rendering. In this variant, the image and text are encoded by the visual encoder and text encoder of the same VLM in their original forms, respectively. The resulting fine-grained visual patches and textual tokens are then concatenated and fed into VRA. As shown in Table 5, although this variant also uses fine-grained multimodal features, it still performs worse than PixelRec. A possible reason is the *modality gap* between visual and textual feature spaces [19]. Even with fine-grained features, separately encoded image and text representations may still be less compatible for downstream fusion [30]. Therefore, this result suggests that another advantage of PixelRec may lie in alleviating modality gap through unified visual encoding. We present this as a supplementary observation rather than a conclusion.

5.5 Visualization Analysis (RQ4)

To provide an intuitive understanding of the fine-grained fusion behavior of PixelRec, we visualize the patch-level importance learned by VRA. Specifically, we select two items for case analysis and map the attention-based importance scores of signal patches back to the corresponding regions of the unified visual-textual input as heatmaps. In the visualization, darker regions indicate higher relative importance when constructing the final item representation.

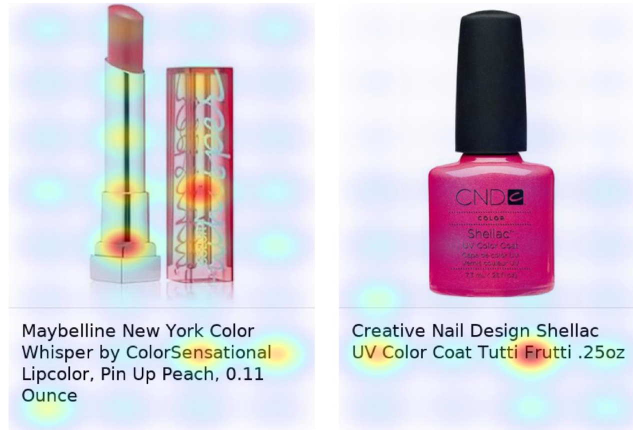

Figure 5: Visualization of PixelRec's signal-patch-level modeling.

From Figure 5, we have two main observations. First, PixelRec performs multimodal fusion at the patch level rather than the modality level, assigning different importance to local regions from both visual and textual content. Second, VRA learns item-specific attention patterns: different items exhibit different high-response regions, indicating that PixelRec can adaptively focus on the most recommendation-relevant patches for each item. Overall, the visualization results further support our motivation that multimodal recommendation should move beyond coarse modality-level fusion toward fine-grained signal-patch modeling.

5.6 Sensitivity Analysis (RQ5)

5.6.1 Compressed Dims of RCC. In PixelRec, the native VLM features are high-dimensional, which leads to substantial GPU memory consumption during downstream training. Therefore, we investigate the trade-off between the RCC compression dimension d_c, recommendation accuracy, and GPU memory usage. As shown in Figure 6, $d_c = 4096$ does not yield the best result on Beauty, while it achieves the best result on Games. Considering both performance and efficiency, we choose $d_c = 1024$ as the default setting. Under this setting, GPU memory usage is reduced from 34GB to 10GB on Beauty and from 32GB to 9GB on Games, corresponding to reductions about 70%. At the same time, 1024 maintains strong recommendation accuracy on both datasets. However, further reducing the compression dimension from 1024 to 256 or 64 causes much larger performance drops. These results show that $d_c = 1024$ provides a favorable trade-off between recommendation accuracy and GPU efficiency. Results on Toys dataset are presented in Appendix.

5.6.2 Patch Sampling Strategy. Since the native VLM produces many signal patches, we investigate three down-sampling strategies: *average_pooling* (AP), *max_pooling* (MP), and *average_sample* (AS). As shown in Table 6, AP performs the best, MP performs the worst, and AS lies in between. This suggests that smooth local

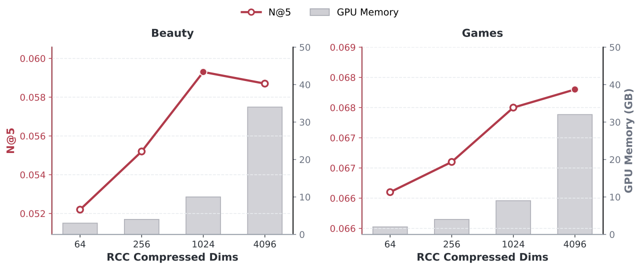

Figure 6: Trade-off between recommendation accuracy and GPU memory consumption under different RCC compressed dimensions. The line denotes N@5 and the bars denote GPU memory usage.

Table 6: Effect of different downsample settings and sampled patch numbers. Results are reported in terms of N@5.

Setting			Size			Beauty	Games	Toys
AP	**AS**	**MP**	**32**	**16**	**8**			
✓			✓			**0.0593**	**0.0664**	**0.0628**
	✓		✓			0.0582	0.0659	0.0621
		✓	✓			0.0573	0.0661	0.0619
✓				✓		0.0571	0.0657	0.0615
✓					✓	0.0556	0.0652	0.0602

Table 7: Effect of different prompt settings. Results are reported in terms of N@5.

Prompt Setting	Beauty	Games	Toys
No Prompt	**0.0593**	**0.0664**	<u>0.0628</u>
Basic	0.0582	0.0655	**0.0630**
Understanding	<u>0.0591</u>	<u>0.0661</u>	0.0625

aggregation is more suitable for recommendation-oriented patch modeling than either local maxima selection or sparse token sampling. We further analyze the effect of the number of downsampled patches under AP, including 32 (8×4), 16 (4×4), and 8 (4×2). Table 6 shows that 32 performs the best, followed by 16, while 8 gives the worst results, indicating that region-level signals should not be downsampled too aggressively.

5.6.3 Prompt Setting. We briefly analyze the effect of prompt settings in Table 7. Specifically, we consider three settings: *No Prompt*, which directly uses the visual input without instructions; *Basic*, which provides a generic instruction; and *Understanding*, which explicitly guides the model to understand the item based on its visual and textual content. The results in Table 7 show that the performance differences among the three prompt settings are small, and *No Prompt* achieves the best results on two of the three datasets. This indicates that PixelRec does not heavily rely on prompt engineering, and can already achieve strong recommendation performance from the visual input alone. Therefore, we consider that the effectiveness of PixelRec mainly comes from its signal-patch-level modeling rather than prompt design. Detailed prompt settings are presented in Appendix.

6 Conclusion

In this paper, we revisit MMSR from the perspective of fusion units. Our analyses suggest that the limitation of existing MMSR methods may stem not only from the fusion function itself, but also from relying on coarse modality-level representations, which do not always translate multimodal inputs into stable complementary gains. Motivated by this, we propose PixelRec, which models each item as a unified visual-textual input and shifts multimodal fusion from modality-level features to fine-grained signal patches. Based on this design, VRA performs recommendation-oriented patch aggregation, while RCC makes high-dimensional patch features more GPU-friendly for downstream training. Experiments on three benchmark datasets show that PixelRec outperforms strong MMSR baselines. Further analyses suggest that these gains are not simply due to using a stronger VLM, but are closely related to the shift from modality-level fusion to signal-patch-level fusion, which also alleviates the modality-compromise phenomenon observed in many existing methods. Overall, our findings highlight that, in MMSR, defining better fusion units can be as important as designing better fusion functions. We hope this perspective could inspire future research on fine-grained multimodal modeling for MMSR.

In future work, we plan to extend the signal-patch-level fusion method of PixelRec to broader recommendation scenarios, including generative recommendation and collaborative filtering.

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